

M002 Extension 3 — Builder Challenge

Topic: Build a Maze + Write the Solution · **Course:** Maze Madness · **Level:** Extension (Intermediate) ·

Time: about 45 minutes

This time **you** make the maze. Build one maze that runs through **6 zones**. Wire it with pistons and doors. Then write the code that drives the agent to the end.

Open **M002 EXT 3 — Cube World** and build inside it. Chat: **l** turns left, **r** turns right, **run** starts your solver, **rl** sends the agent back to start.

Your maze needs at least:

- **2 pistons** (each opens a wall when powered)
- **2 doors** (the agent opens them with `agent.interact(...)`)
- **4 AND** conditions in your code (`... and ...`)
- **1 OR** condition in your code (`... or ...`)

1 · Plan Your 6 Zones

Your maze runs through **6 zones** in a row. The agent finishes one and walks into the next. Give each zone one job: a turn, a piston, or a door.

Zone 1: ...

Zone 2: ...

Zone 3: ...

Zone 4: ...

Zone 5: ...

Zone 6: ...

Which 2 zones get a piston, and what does each open? Which 2 get a door?

2 · Build the Maze

Build it in the world. Walk the path yourself first — make sure a person can reach the end. Tick each part.

- ☐ Zone 1 ☐ Zone 2 ☐ Zone 3 ☐ Zone 4 ☐ Zone 5 ☐ Zone 6
- ☐ Piston #1 tested ☐ Piston #2 tested
- ☐ Door #1 ☐ Door #2
- ☐ A clear START block and GOAL block

What broke while you built it? What did you change?

3 · Write the Solution

Write the solver. Use a `while True` loop so it checks every step. Your code needs **4 AND** and **1 OR**.

```
while True:
    agent.move(FORWARD, 1)

    # turn only when BOTH are true (AND)
    if agent.detect(BLOCK, LEFT) and agent.detect(REDSTONE, AHEAD):
        agent.turn(LEFT_TURN)

    # a door OR a piston gate (OR)
    elif agent.detect(BLOCK, AHEAD) or agent.detect(REDSTONE, RIGHT):
        agent.interact(AHEAD)
        agent.move(FORWARD, 1)
```

Write your own checks for your maze. Add more `if` / `elif` until all 6 zones are covered. Mark each one `# AND` or `# OR`.

How many AND checks did you write? How many OR?

4 · Find the Bug

This code should turn left at a junction. It turns at the wrong time.

```
if agent.detect(BLOCK, LEFT) or agent.detect(REDSTONE, AHEAD):  
    agent.turn(LEFT_TURN)
```

The agent should turn only when **both** are true. Write the fixed line.

5 · Test, Finish, and Show 📷

Type `run` and watch. It will get stuck — that is normal. Find where it stops, fix the code or the maze, run again. Use `r1` to send it back. Keep going until the agent reaches the **GOAL** on its own.

Send a photo OR video of the agent at the goal. Then write 3 or 4 sentences with these starters.

My maze runs through 6 zones. The trickiest zone was ...

I put my pistons in ... and my doors in ...

I used an AND at ... to make the agent ...

I needed an OR at ... because ...

Plan your phone video. Take a video on your phone (or a parent's phone). Show off your maze like a game designer. Write what you will say — you can read from it while filming.

1. Show the whole maze and point out the 6 zones.
2. Show each piston and each door, and say what it does.
3. Read one AND check out loud and say what it decides.
4. Type **run** and let the agent reach the goal.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.